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Belt-fed rotary gun

Abstract

A belt-fed Gatling-style gun is provided. In some embodiments, the belt-feed system allows a plurality of individually separable belt links to be connected in series to feed cartridges to a belt-fed rotary gun. Each individual belt link may be removably connected to a series of belt links to facilitate ease of loading and making belts of ammunition of any length.

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Background/Summary

RELATED APPLICATION (1) This application claims the benefit of U.S. Provisional Application Ser. No. 63/423,240 filed Nov. 7, 2022 for a “BELT-FED ROTARY GUN,” which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD
(1) This disclosure generally relates to Gatling-style guns. More specifically, this disclosure relates to a Gatling-style gun that is belt-fed.
BACKGROUND
(2) Rotary cannons, rotary autocannons, rotary guns, and Gatling-style guns, are a type of firearm with a rotating barrel assembly. The barrels fire as the barrel assembly rotates. In some cases, the barrels may be externally driven via an electric motor; some rotary guns are driven via recoil or gas impulse from spent ammunition; some guns are driven with a manual crank.
SUMMARY
(3) According to one aspect, this disclosure provides a belt-feed assembly for feeding cartridges into the chambers of a rotary gun. In some embodiments, this disclosure includes a belt-feed assembly, which could include a plurality of separable belt links. In some cases, each of the separable belt links includes a cartridge holder configured to hold a cartridge, a belt tooth configured to engage with the gun barrel system, and a feed channel configured to accept a feed ram. Depending on the circumstances, the plurality of separable belt links could be made from a belt material with one or more of plastics, natural rubbers, and/or synthetic rubbers.
(4) In some embodiments, the feed ram may engage a cartridge in the cartridge holder, thereby

forcing the cartridge into a gun barrel system. For example, the cartridge holder may have a variable size cartridge holder configured to accept a wide variety of different caliber munitions, including, but not limited to, 9 mm, 0.22, 0.223, 5.56 mm, and 7.62 mm.
(5) In some cases, the separable belt links could have a first portion and a second portion. The first portion and the second portion may be configured such that the feed channel is formed between the first portion and the second portion of the separable belt links.
(6) Embodiments are contemplated in which the separable belt links include a belt link connector. For example, the belt link connector of the separable belt links could have a hook portion and a rail portion. The hook portion of the separable belt links may be configured to engage with the rail portion of the separable belt links such that a continuous belt is formed by joining multiple separable belt links together. The belt tooth could be configured to engage with a belt tooth acceptor on the gun barrel system such that the belt tooth on the separable belt links is drawn into the gun barrel system sequentially.
(7) Depending on the circumstances, the system might have a lock door. For example, the lock door could include a feed tray configured to accept the separable belt links.
(8) According to another aspect, this disclosure provides a system for feeding cartridges to a rotary gun. In some embodiments, the system may have a delivery vehicle with a plurality of separable belt links. In some cases, each of the plurality of separable belt links has a cartridge holder configured to hold a cartridge. Each of the plurality of separable belt links may have a belt tooth configured to engage with the gun barrel system. Additionally, in some embodiments, each of the plurality of separable belt links includes a feed channel configured to accept a feed ram. During operation, the delivery vehicle could orient cartridges with respect to the gun barrel system, feed cartridges by being pulled through the gun barrel system via the belt tooth on each belt link, load a cartridge into the gun barrel system by injecting it in the gun barrel system via a feed ram, and eject from the gun barrel system.
(9) According to a further aspect, this disclosure provides a method for belt-feeding a rotary gun. In some embodiments, the method includes orienting a cartridge within a reusable modular belt. The reusable modular belt is fed to a rotary gun via a series of belt teeth on the reusable modular belt. For example, the belt teeth on the reusable modular belt could be configured to slot into a gun barrel system such that they are pulled through the gun barrel system. The method may include the step of loading the cartridge into the gun barrel system via a feed ram. In some cases, the feed ram is configured to pass through a central channel in the reusable modular belt and disengage the cartridge from the reusable modular belt. The method may also include injecting the cartridge into the gun barrel system, and ejecting the reusable modular belt from the rotary gun.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The present disclosure will be described hereafter with reference to the attached drawings which are given as non-limiting examples only, in which:
(2) FIG. **1** is a perspective view of an example belt-fed rotary gun according to an embodiment of this disclosure;
(3) FIG. **2** is a perspective view of two example belt links that could be part of a belt assembly for feeding cartridges into the belt-fed rotary gun of FIG. **1** according to an embodiment of this disclosure;
(4) FIG. **3** is a perspective view of the example belt links of FIG. **2** connected together according to an embodiment of this disclosure;
(5) FIG. **4** is a perspective view of a single belt link that could form a portion of a belt assembly according to an embodiment of this disclosure;

(6) FIG. **5** is a perspective view of an example belt assembly within the belt-fed rotary gun of FIG. **1** as it is loading a cartridge in a first position;
(7) FIG. **6** is a perspective view of the example belt assembly within the belt-fed rotary gun of FIG. **1** as it is loading a cartridge in a second position;
(8) FIG. **7** is a perspective view of an example lock door within the belt-fed rotary gun of FIG. **1** according to an embodiment of this disclosure;
(9) FIG. **8** is a cross-sectional view of the belt assembly within the belt-fed rotary gun of FIG. **1** as it is loading a cartridge in a first position;
(10) FIG. **9** is a detailed view of the cross-section shown in FIG. **8** as the gun loads a cartridge in a second position; and
(11) FIG. **10** is a detailed view of the cross-section shown in FIG. **9** as the gun loads a cartridge in a third position.
(12) Corresponding reference characters indicate corresponding parts throughout the several views. The components in the figures are not necessarily to scale, emphasis instead being placed upon illustrating the principals of the invention. The exemplification set out herein illustrates embodiments of the invention, and such exemplification is not to be construed as limiting the scope of the invention in any manner.

DETAILED DESCRIPTION OF THE DRAWINGS

(13) While the concepts of the present disclosure are susceptible to various modifications and alternative forms, specific exemplary embodiments thereof have been shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that there is no intent to limit the concepts of the present disclosure to the particular forms disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the disclosure.
(14) References in the specification to “one embodiment,” “an embodiment,” “an illustrative embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may or may not necessarily include that particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to effect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. Additionally, it should be appreciated that items included in a list in the form of “at least one A, B, and C” can mean (A); (B); (C); (A and B); (A and C); (B and C); or (A, B, and C). Similarly, items listed in the form of “at least one of A, B, or C” can mean (A); (B); (C); (A and B); (A and C); (B and C); or (A, B, and C).
(15) The present disclosure is directed towards a belt-fed rotary gun. In some embodiments, this disclosure provides a technical advantage of having an adjustable length belt assembly. This allows the user to determine how many cartridges to feed the gun. For example, the belt assembly could be adjusted between X links and Y links to adjust the number of cartridges fed into the gun from X cartridges to Y cartridges, similar to how magazines have different capacities. Another technical advantage, in some embodiments, is the direct feed of the belt assembly into the gun's chambers; instead of the belt placing the cartridges into another feed mechanism that inserts the cartridge into the chamber, the belt assembly in some embodiments directly deposits cartridges into the chambers for firing. For example, each link in the belt assembly could drop a cartridge via gravity feed directly into respective chambers of the gun as the belt is fed through the gun.
(16) Turning to FIG. **1**, an example embodiment of a rotary gun **100** is shown. As shown, the rotary gun **100** has a gun barrel system **110**, a belt-feed assembly **120**, a gun lock system **130**, and a gun drive system **140**.
(17) In the embodiment shown, the gun barrel system **110** includes a plurality of barrels **111** arranged in a cyclic array. As the barrels **111** rotate, the rotary gun **100** fires providing saturational

directional fire. As shown, the belt-feed assembly **120** is formed from a plurality of individual belt links **121**. The individual belt links **121** may be joined together to form a continuous belt with any number of individual belt links **121**. Each belt link **121** has a cartridge holder **124** built into the belt link **121**. The cartridge holder **124** is configured to accept and retain a cartridge **122** which the belt-feed assembly **120** cycles through the rotary gun **100**.

(18) As shown, the gun lock system **130** has a lock door **131** and a gun firing assembly **132**. The lock door **131** covers the gun firing assembly **132** during operation and prevents dirt and grime from agglomerating in the gun lock system **130**. Additionally, the lock door **131**, in some embodiments, may be removable such that jams in the gun firing assembly **132** during operation may be cleared quickly. The belt-feed assembly **120** is cycled through the gun lock system **130** to load the rotary gun **100** during operation. The gun firing assembly **132** is the location where the cartridge **122** is disengaged from the cartridge holder **124** in the belt link **121** and deposited into a chamber **135** (FIG. **10**) of the gun lock system **130**. Each of the plurality of barrels **111** has a chamber **135**. In the embodiment shown, the gun lock system **130** also has a feed ram **134** which disengages the cartridge **122** from the belt link **134** and effectively loads the rotary gun **100**.

(19) The gun drive system **140** drives the rotation of the barrels **111** during operation. In some embodiments, the gun drive system **140** includes a handle **142** and mounting holes **143** that allow the rotary gun **100** to be mounted and used by an operator. As shown, the gun drive system **140** also has a drive unit **141**. The drive unit **141** may be gas or recoil actuated, electrically actuated via a motor (not shown), or manually actuated via a crank handle (not shown).

(20) Turning next to FIG. **2**, an example embodiment of a belt-feed assembly **120** is shown. As shown, the belt-feed assembly **120** includes a plurality of connected belt links **121**. In some embodiments, the connected belt links **121** further comprise a first portion **121***a* and a second portion **121***b*. The first portion **121***a* and second portion **121***b* are substantially parallel. Any number of individual belt links **121** may be connected together to form a belt-feed assembly **120** with a belt of any length. For example, individual belt links could have a cartridge holder **124** configured to accept a cartridge **122** of varying size. The cartridge holder **124** may be in the shape of a semicircular detent. Any number of different size cartridges may be used, including, but not limited to, 9 mm, 0.22, 0.223, 5.56 mm, and 7.62 mm.

(21) In the embodiment shown, each belt link **121** includes at least one belt tooth **127**. The belt tooth **127** engages with the gun barrel system **110**. The belt tooth **127** of a first belt link **121** is spatially distanced from the belt tooth **127** of a second belt link **121** such that they match with a plurality of indents on the gun barrel system **110**. The gun barrel system **110** draws the belt links **121** through the rotary gun **100** during operation, effectively allowing for a continuous influx of cartridges **122** to the rotary gun **100**, which in turns provides for continuous saturational directional fire from the rotary gun **100**. As shown, the belt links **121** also have a feed channel **125** which allows for the feed ram **134** to disengage the cartridge **122** from the belt link **121** during operation. The feed channel **125** is formed between the first portion **121***a* and second portion **121***b* on the belt link **121**. In some embodiments, the belt link **121** also has a belt connector **123** which allows multiple belt links **121** to be joined together to form belts of varying lengths depending on the needs of the operator. The belt connector **123** further includes a belt hook **126** which is designed to slot on to a belt rail **129** on an adjoining belt link **121** in a rotatable manner. The belt rail **129**, in some embodiments, is substantially cylindrical. This linkage allows belts of varying lengths to be quickly and easily assembled and disassembled, and does not require any specialized tools.

(22) Turning next to FIG. **3**, the linkage between two individual belt links **121** is shown. The belt hook **126** of the belt connector **123** is shown slotted over the belt rail **129** of the adjoining belt link **121**. FIG. **4** further shows a single belt link according to this embodiment in detail.

(23) FIG. **5** shows the rotary gun **100**. The belt links **121** containing the cartridge **122** are in a first position where the cartridge **121** in the belt link **121** has not yet been disengaged from the belt link **121** via the feed ram **134** and inserted into the gun firing assembly **132**.

(24) FIG. **6** shows the rotary gun **100** in a second position where the cartridge **122** has been disengaged from the belt link **121** by the feed ram **134** and inserted into the chamber **135** within the gun firing assembly **132**. In the second position, the barrel **111** of the gun barrel system **110** is in a position such that it is ready to be fired.
(25) Turning next to FIG. **7**, a lock door **131** is shown. The lock door **131** covers the gun firing assembly **132** and each chamber **135** during operation of the rotary gun **100**. The lock door **131** covers the gun firing assembly **132** during operation and prevents dirt and grime from agglomerating in the gun lock system **130**. The lock door **131**, in some embodiments, is removable such that any jams during operation of the rotary gun **100** may be cleared quickly and efficiently on the field. The lock door **131** further comprises the feed ram **134** which disengages the cartridge from the belt link **121** and forces it into the gun lock system **130**. The lock door **131** is connected to the gun lock system **130** via a feed tray **133**. The feed tray **133** guides the belt links **121** and cartridges **122** into the gun firing assembly **132** in preparation for loading the rotary gun **100**.
(26) FIGS. **8-10** are progressive views showing the belt-feed assembly **120** depositing a cartridge into a chamber via the gun firing assembly **132** according to an embodiment. FIG. **8** shows the loading process in a first position. The belt links **121** are fed into the gun lock system **130** via the feed tray **133**. The belt links **121** are joined together via the belt hook **126** and belt rail **129** forming a continuous belt which allows for consistent and continuous feed to the gun lock system **130**. In the first position, the cartridge **122** is still engaged to the belt link **121** via the cartridge holder **124**. The feed ram **134** has not disengaged the cartridge **122** and deposited it into the gun firing assembly **132** and chamber **135** in the first position. The belt tooth **127** is engaged to the gun barrel system **110** via a belt tooth acceptor **128**. The belt tooth acceptor **128** pulls the belt links **121** via the belt tooth **127** located on each belt link **121** into the gun barrel system **110** and gun lock system **130**. The continuous feed of cartridges via the belt links **121** and this mechanism allow for continuous saturational directional fire of the rotary gun **100** during operation.
(27) FIG. **9** shows the loading process in a second position. The belt link **121**, via the belt tooth **127** being pulled forward into the gun lock system **130** by the belt tooth acceptor **128** on the gun barrel system, has advanced forward into the gun firing assembly **132**. The cartridge **122** contained in the cartridge holder **124** has engaged the feed ram **134**. The feed ram **134** has begun to disengage the cartridge **122** from the cartridge holder **124** on the belt link **121**. The cartridge **122** has started to be forced into the chamber **135** in the gun firing mechanism **132** in the first position. The belt-feed assembly **120** is pulled into the gun lock system **130** by the belt tooth **127** engaging with the belt tooth acceptor **128**.
(28) FIG. **10** shows the loading process in a third position. In the third position, the feed ram **134** has disengaged the cartridge **122** from the cartridge holder **124** on the belt link **121**. The cartridge **122** has been injected into the gun lock system **130** via the gun firing assembly **132** and chamber **135** within the respective barrel **111** being loaded. The lock door covers the gun lock system **130** during operation. Once the cartridge **122** has been fully disengaged from the belt link **121** and injected into the gun lock system **130**, the barrel **111** rotates to the firing position and the cartridge **122** is fired. The barrel **111** continues rotating and ejects the cartridge **122** after firing as well as ejects the belt link **121** after the cartridge **122** has been disengaged from the belt link **121**. The belt link **121** after being ejected from the rotary gun **100** can be reused. This simplified loading system for injecting cartridges **122** into the gun lock system **130** reduces the amount of maintenance needed because there are fewer moving parts in the gun lock system **130** and belt-feed assembly **120** than in the prior art. It is advantageous to have easier serviceability with fewer parts to ensure the rotary gun **100** can be easily maintained and serviced in the field.
(29) It is to be understood that the invention is not limited to the exact details of construction, operation, exact materials or embodiments shown and described, as obvious modifications and equivalents will be apparent to one skilled in the art. While the specific embodiments have been illustrated and described, numerous modifications come to mind without significantly departing

from the spirit of the invention, and the scope of protection is only limited by the scope of the accompanying Claims.

Claims

1. A rotary gun comprising: belt-feed assembly configured to feed ammunition through the rotary gun, the belt feed assembly comprising: (i) a first belt link comprising a first elongated side and a second elongated side, wherein the first elongated side of the first belt link is substantially parallel to the second elongated side of the first belt link, wherein the first elongated side of the first belt link further comprises a first semicircular detent configured to secure a cartridge, wherein the second elongated side of the first belt link further comprises a second semicircular detent configured to secure a cartridge, wherein the first elongated side of the first belt link is connected to the second elongated side of the first belt link with a substantially cylindrical cross member, wherein the first belt link comprises a hook opposite the substantially cylindrical cross member, (ii) a second belt link comprising a first elongated side and a second elongated side, wherein the first elongated side of the second belt link is substantially parallel to the second elongated side of the second belt link, wherein the first elongated side of the second belt link further comprises a first semicircular detent configured to secure a cartridge, wherein the second elongated side of the second belt link further comprises a second semicircular detent configured to secure a cartridge, wherein the first elongated side of the second belt link is connected to the second elongated side of the second belt link with a substantially cylindrical cross member, wherein the second belt link comprises a hook opposite the substantially cylindrical cross member, wherein the hook on the second belt link is configured to be attached to the substantially cylindrical cross member of the first belt link; and a gun firing assembly including a feed ram configured to sequentially disengage cartridges fed into the rotary gun by the belt-feed assembly.
2. The rotary gun of claim 1, wherein the first belt link and second belt link further comprise a plurality of belt teeth.
3. The rotary gun of claim 2, wherein the plurality of belt teeth of the first belt link and the second belt link are spatially arranged to engage with a plurality of indents in a barrel system of the rotary gun.
4. The rotary gun of claim 3, wherein the plurality of indents in a barrel system of a gun are configured to pull the first belt link and second belt link sequentially into the rotary gun.
5. The rotary gun of claim 1, wherein the first semicircular detent and second semicircular detent of the first belt link further comprise variable size semicircular detents configured to accept a variety of different munitions.
6. The rotary gun of claim 1, wherein the first semicircular detent and second semicircular detent of the second belt link further comprise variable size semicircular detents configured to accept a variety of different munitions.
7. The rotary gun of claim 1, wherein the first belt link and second belt link further comprise a belt material, and wherein the belt material is selected from one or more of: plastics, natural rubbers, and/or synthetic rubbers.
8. The rotary gun of claim 1, wherein the first elongated side and the second elongated side are configured to flex along an axis transverse to an longitudinal axis of the first elongated side and the second elongated side.
9. The rotary gun of claim 8, wherein the feed ram engages a cartridge in the first semicircular detent and second semicircular detent of the first belt link forcing the cartridge into a gun barrel system.
10. The rotary gun of claim 8, wherein the feed ram engages a cartridge in the first semicircular detent and second semicircular detent of the second belt link forcing the cartridge into a gun barrel system.

11. The rotary gun of claim 1, wherein the first elongated side and the second elongated side of the first belt link are configured to form a feed channel between the first elongated side and the second elongated side of the first belt link.

12. The rotary gun of claim 1, wherein the first elongated side and the second elongated side of the second belt link are configured to form a feed channel between the first elongated side and the second elongated side of the second belt link.

13. The rotary gun of claim 1, wherein the rotary gun further comprises a lock door.

14. The rotary gun of claim 13, wherein the lock door further comprises a removable lock door.

15. The rotary gun of claim 14, wherein the lock door further comprises a feed tray configured to guide the first belt link and second belt link into a gun barrel system.

16. A method for belt-feeding a rotary gun, the method comprising: orienting a cartridge within a reusable modular belt link on a reusable modular belt, feeding the reusable modular belt to a rotary gun via a series of belt teeth on the reusable modular belt, wherein the belt teeth on the reusable modular belt are configured to slot into a gun barrel system such that the series of belt teeth are pulled through the gun barrel system, loading the cartridge into the gun barrel system via a feed ram, wherein the feed ram is configured to pass through a central channel in the reusable modular belt and disengage the cartridge from the reusable modular belt, and further inject said cartridge into the gun barrel system, and ejecting the reusable modular belt from the rotary gun.

17. The method of claim 16, further comprising connecting a plurality of reusable modular belt links in series to form the reusable modular belt.

18. The method of claim 16, further comprising reusing the plurality of reusable modular belt links in series to form the reusable modular belt.